

Advanced Economic Statistics

Course Objectives

The aim of this course is to prepare participants to evaluate and undertake statistical analysis. The objectives of this course are to enable the participants to be familiar with a wide range of statistical techniques and be able to generate and interpret basic statistical analysis.

Course Contents

1. Introduction

Introduction of the subject, scope & Limitations. Methods of Data Collection; Census and sample survey, types of Data; Primary & Secondary Data. Data Management; Approximation, rounding Cleaning & Tabulation.

2. Frequency Distribution

Types of Variables, Types of Grouping Classes & Class Interval, Patterns of Frequency Distribution, Cumulative Frequency Distribution, Bivariate Frequency Distribution. Diagrammatic & Graphical representation, Bar diagram, Pie Diagram, Histogram, Frequency Curve & Ogive.

3. Measures of Central Tendency

Averages. Arithmetic Mean, Median, Mode. Relationship between Mean Median & Mode. Geometric & Harmonic mean, Partition Values; Quartiles, Deciles & Percentiles.

4. Measures of Dispersion

Range, Mean Deviation, Standard Deviation, Quartile Deviation, Relative measures of dispersion, Coefficient of Variation.

5. Moments, Skewness & Kurtosis

Absolute Moments, Raw Moments, True Moments, Skewness & Kurtosis.

6. Correlation & Regression Analysis

Simple Correlation, Rank Correlation, Simple Linear Regression Equation of Y on X, Significance of Regression Coefficient, Concept of Elasticity through Regression Coefficient.

7. Index Number

Simple Indices, Aggregative Indices, Weighted Aggregative Indices, Consolidation of Series & Shifting of Base.

8-Probability

Introduction to Inferential Statistics, Basic concepts regarding probability. Definitions of Probability, Theorems of Total and Compound Probability, Generalized Theorem of Total Probability, Conditional Probability, Bay's theorems.

9. Probability Distributions

Random Variables and their Probability distributions, Binomial Distribution, Poisson distribution, Normal Distribution, Distribution of Sample Mean, Central Limit Theorem.

10. Tests of Hypotheses:

Null and Alternate Hypotheses, Critical and Non-critical regions. Testing the significance of an observed mean, Testing the significance of an observed Variance.

Recommended Books

1. Weiss, N. A. (2017). *Introductory Statistics*. Global Edition. Pearson Education Limited.
2. Anderson D. R., Sweeney, D. J., and Williams, T. A. (2002). *Statistics for Business and Economics*. South-Western, Thomson Learning.
3. Keller G. and Warrack (2000). *Statistics for Management and Economics*. New York: Duxbury.
4. Berenson, M. L., Levine, D. M., and Krehbiel, T. C. (2012). *Basic Business Statistics: Concepts and Applications*. 12th Edition, Prentice-Hall.
5. Wonnacott, T., and Wonnacott, R. (1990). *Introductory Statistics for Business and Economics*. John Wiley & Sons, Inc.

Note: 1. Duration of course will be 10 weeks.

2. Total 10 lectures of two hours duration will be delivered.

3. Class time will be from 7:00 pm to 9:00 pm, once a week.

4. Atleast 75% attendance is required to get a certificate.